

Asterisms

A web project for learning to read the night sky: puzzle games on a real star catalogue, and a location-aware sky viewer built on the same engine.

Role: solo builder

Private full-stack product

Live: asterisms.space

Next.js 16.3.4

React 19.2.8

TypeScript 5.9.3



Completed puzzle reveal captured from the current product.

162

generated patterns

9,042

catalogue stars

347

passing tests

SYSTEM SUMMARY

Licensed astronomy
sources



Sync, build, validate
pipeline



Compact generated
catalogue

Browser gameplay
and real-sky viewer



Next.js API routes



Restricted database
runtime access

KEY ENGINEERING DECISIONS

- Joined multiple astronomy catalogues so puzzle shapes use real star coordinates, magnitudes, and identifiers instead of hand-positioned points.
- Projected RA/Dec with a gnomonic tangent-plane projection, then rotated and scaled patterns without distorting their geometry.
- Made Daily deterministic from UTC date so the puzzle remains playable even if score persistence is unavailable.
- Separated realistic star rendering from touch-friendly SVG hit targets for mobile and desktop play.
- Built a real-sky layer (Tonight, Seek from 1900 to now, Guided Find, events calendar) on the same catalogue, keeping precise location out of storage, analytics, servers and share links.

LIMITATIONS

- Not a professional astronomy simulator. Visibility is geometric only: no weather, light pollution, terrain or refraction.
- No accounts; score identity is anonymous and device-local.
- Score validation is a plausibility filter, not full anti-cheat.

QUALITY + RELIABILITY

- Current validation passes: data validation, tests, typecheck, lint, and production build.
- Generated data validation reports zero warnings and zero hard failures.
- Database-backed routes degrade gracefully; core gameplay catalogue is static app data.
- CI runs validation, tests, typecheck, lint, and build.
- Astronomy output spot-checked against NASA/JPL Horizons and USNO for five observers and dates.

WHAT IT DEMONSTRATES

Asterisms demonstrates domain learning, data pipeline design, interactive rendering, product UX, API design, database-backed leaderboards, graceful degradation, and validation discipline across a real full-stack project.

- Mythology and broader sky-culture story text are deferred pending reliable sources.

Figures verified 2 October 2026. Public-safe case study. No app source code, credentials, private repository links, account identifiers, or internal deployment identifiers included.